Statistics & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Permutations**

How many ways can 5 distinct books be arranged on a shelf?

$5! = 120$.

2 **[Foundational]** • **Target: Permutations**

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 12 members?

${}_{12}P_3 = 12 \times 11 \times 10 = 1320$.

3 **[Foundational]** • **Target: Permutations**

How many distinct arrangements are there of the letters in the word APPLE?

There are 5 letters with the letter P repeated twice: $\frac{5!}{2!} = \frac{120}{2} = 60$.

4 **[Foundational]** • **Target: Combinations**

How many ways can a committee of 4 be chosen from 11 people?

$\binom{11}{4} = \frac{11!}{4!\,7!} = 330$.

5 **[Foundational]** • **Target: Combinations**

A pizza shop offers 8 toppings. How many different 3-topping pizzas can be made?

$\binom{8}{3} = \frac{8 \times 7 \times 6}{6} = 56$.

6 **[Foundational]** • **Target: Combinations**

From a group of 6 men and 5 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?

$\binom{6}{2}\binom{5}{2} = 15 \times 10 = 150$.

7 **[Foundational]** • **Target: Probability rules**

A fair die is rolled. Find $P(\text{even or greater than } 4)$.

Even $= \{2, 4, 6\}$, greater than $4 = \{5, 6\}$. Union $= \{2, 4, 5, 6\}$. $P = \frac{4}{6} = \frac{2}{3}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Probability rules

$P(A) = 0.45$ and $P(B) = 0.3$, and A and B are mutually exclusive. Find $P(A \text{ or } B)$.

$P(A \text{ or } B) = 0.45 + 0.3 = 0.75$.

9 [Intermediate] • Target: Probability rules

A card is drawn from a standard 52-card deck. Find $P(\text{not a face card})$.

There are 12 face cards, so 40 are not face cards. $P = \frac{40}{52} = \frac{10}{13}$.

10 [Intermediate] • Target: Conditional probability and independence

$P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A and B independent?

$P(A)P(B) = 0.5 \times 0.4 = 0.2 = P(A \cap B)$. Yes, A and B are independent.

11 [Intermediate] • Target: Conditional probability and independence

In a class, 60% of students play sports, and 25% of all students both play sports and are honor students. Find $P(\text{honor student} \mid \text{plays sports})$.

$P(\text{honor} \mid \text{sports}) = \frac{0.25}{0.6} \approx 0.417$, or about 41.7%.

12 [Intermediate] • Target: Conditional probability and independence

Events C and D are independent, with $P(C) = 0.7$ and $P(D) = 0.6$. Find $P(C \cup D)$.

$P(C \cap D) = 0.7 \times 0.6 = 0.42$. $P(C \cup D) = 0.7 + 0.6 - 0.42 = 0.88$.

13 [Intermediate] • Target: Measures of center and spread

Find the mean and median of the data set $12, 15, 18, 20, 25$.

Mean $= \frac{12 + 15 + 18 + 20 + 25}{5} = \frac{90}{5} = 18$. Median (middle value, already sorted) $= 18$.

14 [Intermediate] • Target: Measures of center and spread

Find the population standard deviation of the data set $4, 8, 6, 5, 3$.

Mean $= \frac{26}{5} = 5.2$. Squared deviations: $1.44, 7.84, 0.64, 0.04, 4.84$, summing to 14.8. Variance $= \frac{14.8}{5} = 2.96$. Standard deviation $= \sqrt{2.96} \approx 1.72$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Data representation • Cross-Lesson: 1.01 Advanced Algebra

The bar chart shows the distribution of grades in a class of 37 students. Find the probability that a randomly selected student earned a grade of A or B, as a percent to the nearest whole percent.

$P(A \text{ or } B) = \frac{8+15}{37} = \frac{23}{37} \approx 0.622$, or about 62%.

16 [Challenging] • Target: Data representation • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Using the same bar chart, find the probability that a randomly selected student scored a C or D grade, as a percent to the nearest whole percent.

$P(C \text{ or } D) = \frac{10+4}{37} = \frac{14}{37} \approx 0.378$, or about 38%.

17 [Challenging] • Target: The binomial theorem • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the coefficient of x^3y^2 in the expansion of $(x+y)^5$.

The term is $\binom{5}{2}x^3y^2$, so the coefficient is $\binom{5}{2} = 10$.

18 [Challenging] • Target: The binomial theorem • Cross-Lesson: 4.01 Lines, Triangles & Polygons

In the expansion of $(2x-1)^4$, find the coefficient of x^2.

The general term is $\binom{4}{k}(2x)^{4-k}(-1)^k$. For x^2, $4-k=2 \Rightarrow k=2$: $\binom{4}{2}(2x)^2(-1)^2 = 6(4x^2) = 24x^2$. The coefficient is 24.

19 [Challenging] • Target: The normal distribution • Cross-Lesson: 5.01 Trigonometry

A data set is normally distributed with mean 100 and standard deviation 15. What percent of values lie between 70 and 130?

70 and 130 are each 2 standard deviations from the mean (100 ± 30). By the 68-95-99.7 rule, about 95% of values lie in this range.

20 [Challenging] • Target: The normal distribution • Cross-Lesson: 6.01 Circles & Coordinate Geometry

For the same distribution (mean 100, standard deviation 15), find the approximate percentage of values greater than 115.

115 is 1 standard deviation above the mean. About 68% of values lie within 1 standard deviation, so 34% lie between the mean and 115. Since 50% lie above the mean, the percentage above 115 is $50\% - 34\% = 16\%$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Permutations

Find the population variance of the data set: $2, 8, 12, 8, 10$.

a. 173/15

b. **56/5 ✓**

c. 677/60

d. 163/15

Mean $= 8$. Variance $= \frac{(2-8)^2, (8-8)^2, (12-8)^2, (8-8)^2, (10-8)^2}{5} = \frac{56}{5} = 56/5$.

22 [Foundational] • Target: Combinations

Find the median of the data set: $9, 4, 9, 2, 7$.

a. 17

b. 24

c. 13

d. **7 ✓**

Sorted: $2, 4, 7, 9, 9$. With 5 values, the median is the middle value: 7.

23 [Foundational] • Target: Probability rules

Find the mean of the data set: $23, 11, 31, 33, 30, 34$.

a. 25

b. **27 ✓**

c. 5

d. 42

Mean $= \frac{23+11+31+33+30+34}{6} = \frac{162}{6} = 27$.

24 [Foundational] • Target: Conditional probability and independence

A data set: 2, 5, 5, 7, 9. The mode is:

a. **5 ✓**

b. 6

c. 7

d. 2

5 appears twice, all others once.

25 [Foundational] • Target: Measures of center and spread

68% of data in a normal distribution falls within $0.5cm$ standard deviations of the mean.

a. 3

b. 2

c. 1 ✓

d. 1.5

Applying Probability & Statistics principles, 68-95-99.7 rule. Therefore, the verified correct option is 1.

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26 [Foundational] • Target: Data representation

95% of data in a normal distribution falls within $0.5cm$ standard deviations.

a. 1

b. 3

c. 2 ✓

d. 1.5

Applying Probability & Statistics principles, 68-95-99.7 rule. Therefore, the verified correct option is 2.

27 [Foundational] • Target: The binomial theorem

The bar chart below shows the number of students who chose different sports as their favorite. What is the total number of students surveyed?

a. 50

b. 60

c. 70

d. **80** ✓

To find the total number of students surveyed, we need to sum the number of students for each sport. From the bar chart: - Soccer: 20 students - Basketball: 15 students - Tennis: 10 students - Swimming: 25 students Total students = 20 + 15 + 10 + 25 = 70 Therefore, the total number of students surveyed is 70.

28 [Foundational] • Target: The normal distribution

The following bar chart shows the number of students who prefer different fruits. What is the total number of students who prefer apples and bananas?

a. 20

b. **25** ✓

c. 30

d. 35

From the bar chart, the number of students who prefer apples is 15. The number of students who prefer bananas is 10. The total number of students who prefer apples and bananas is the sum of these two values: 15 + 10 = 25.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: quality control sampling

The bar chart below shows the number of students who achieved different scores on a recent quiz. What is the total number of students who scored 8 or 10?

a. 25

b. **30 ✓**

c. 35

d. 40

The bar chart shows that 15 students scored 8 and 20 students scored 10. Therefore, the total number of students who scored 8 or 10 is $15 + 20 = 35$.

30 [Intermediate] • Target: Permutations

The table shows exam marks for 20 students. Find the mean and identify the modal class.

a. **Mean ≈ 65, Modal class 60–69 ✓**

b. Mean ≈ 68, Modal class 70–79

c. Mean ≈ 60, Modal class 50–59

d. Mean ≈ 72, Modal class 60–69

Mid-points: 44.5(2)+54.5(4)+64.5(7)+74.5(5)+84.5(2) = 89+218+451.5+372.5+169 = 1300; mean = 1300/20 = 65. Modal class = 60–69 (highest frequency).

31 [Intermediate] • Target: Combinations

In a group of 20 students, 9 study French, and 1 study both French and Spanish. If a student is chosen at random from those studying French, what is the probability they also study Spanish?

a. **1/9 ✓**

b. 7/36

c. 19/90

d. 5/18

$P(\text{Spanish} \mid \text{French}) = \frac{P(\text{both})}{P(\text{French})} = \frac{1/20}{9/20} = \frac{1}{9} = 1/9$.

32 [Intermediate] • Target: Probability rules

A bag has 23 marbles, of which 2 are red. What is the probability of drawing a red marble?

a. 43/230

b. **2/23 ✓**

c. 89/230

d. 33/115

Applying Statistics & Probability principles, $P(\text{red}) = \frac{2}{23}$. Therefore, the verified correct option is 2/23.

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33 [Intermediate] • Target: Conditional probability and independence

In how many ways can 2 of 7 books on a shelf be arranged?

a. 82

b. 33

c. **42** ✓

d. 341

Applying Statistics & Probability principles, $7P2 = \frac{7!}{(7-2)!} = 42$. Therefore, the verified correct option is 42.

34 [Intermediate] • Target: Measures of center and spread

In how many ways can 3 of 4 photos in an album be arranged?

a. 132

b. 69

c. 154

d. **24** ✓

Applying Statistics & Probability principles, $4P3 = \frac{4!}{(4-3)!} = 24$. Therefore, the verified correct option is 24.

35 [Intermediate] • Target: Data representation

In how many ways can 2 of 4 runners in a race (1st through r-th place) be arranged?

a. 296

b. 112

c. **12** ✓

d. 270

Applying Statistics & Probability principles, $4P2 = \frac{4!}{(4-2)!} = 12$. Therefore, the verified correct option is 12.

36 [Intermediate] • Target: The binomial theorem

A basketball player makes a free throw with probability $2/3$. If she takes 2 free throws, what is the probability she makes at least one?

a. 53/90

b. 55/72

c. **8/9** ✓

d. 35/36

$P(\text{at least one}) = 1 - P(\text{none}) = 1 - (1/3)^2 = 1 - 1/9 = 8/9$.

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37 [Intermediate] • Target: The normal distribution

A committee of 5 people is chosen from 9 people, and then 4 of those 5 are assigned distinct roles (e.g., chair, secretary, ...). How many ways can this be done?

a. 2696

b. **15120 ✓**

c. 396

d. 4484

Choose the committee: 95. Assign roles: $5P4$. Total $= 95P4 = 126120 = 15120$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: quality control sampling • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Events A and B are mutually exclusive, with $P(A) = 2/7$ and $P(B) = 3/8$. Find $P(A \text{ or } B)$.

a. **37/56 ✓**

b. 269/280

c. 55/168

d. 83/168

Mutually exclusive events: $P(A \text{ or } B) = P(A) + P(B) = 2/7 + 3/8 = 37/56$.

39 [Challenging] • Target: Permutations • Cross-Lesson: 5.01 Trigonometry

The scatter plot shows the relationship between temperature (°C) and ice-cream sales (hundreds). Estimate the sales when temperature is 30°C.

a. About 300

b. About 400

c. **About 500 ✓**

d. About 600

Reading the trend line at 30°C gives approximately 500 sales.

40 [Challenging] • Target: Combinations • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Which measure is most affected by an outlier?

a. Median

b. Mode

c. **Mean ✓**

d. Range (equally)

The mean changes when an extreme value is added.

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41 [Challenging] • Target: Probability rules • Cross-Lesson: 7.01 Solid Geometry & Similarity

A team of 4 is chosen from 9 players. How many different teams are possible?

a. 120

b. **126 ✓**

c. 3024

d. 84

$\binom{9}{4} = \frac{9!}{4! \cdot 5!} = 126.$

42 [Challenging] • Target: Conditional probability and independence • Cross-Lesson: 1.01 Advanced Algebra

A bag has 10 marbles, of which 4 are blue. Two marbles are drawn without replacement. What is the probability both are blue?

a. 1/30

b. 1/20

c. **2/15 ✓**

d. 7/30

$P = \frac{4}{10} \times \frac{3}{9} = 2/15.$

43 [Challenging] • Target: Measures of center and spread • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In how many ways can 3 of 9 students in a row be arranged?

a. 162

b. 531

c. 352

d. **504 ✓**

$9P3 = \frac{9!}{(9-3)!} = 504.$

44 [Challenging] • Target: Data representation • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A committee of 6 people is chosen from a group of 9 people. How many different committees are possible?

a. **84 ✓**

b. 142

c. 111

d. 134

$96 = 9!\,6!\,(9-6)! = 84.$

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45 [Challenging] • Target: The binomial theorem • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Show that the binomial distribution approaches Poisson as $n \to \infty$, $p \to 0$, $np = \lambda$ fixed.

a. Requires $n > 100$

b. Not true

c. $\binom{n}{k} p^k (1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!}$ **— use** $(1-\lambda/n)^n \to e^{-\lambda}$ ✓

d. Only for $k = 0$

Applying Probability & Statistics principles, Poisson limit theorem. Therefore, the verified correct option is $\binom{n}{k} p^k (1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!}$ — use $(1-\lambda/n)^n \to e^{-\lambda}$.

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